BIS30903 : COMMUNICATION AND NETWORK SECURITY

NETWORK SECURITY PROPOSAL
FOR
XYZ Corp. (ThreatHunter Enterprise)

PREPARED BY:

Group 20

GROUP MEMBERS:

1	ALKHATTAB SHEHABADDIN ABDULLAH ABDULWADOD AL-DUBAI	AI230209
2	MOHAMMAD SYAZWAN BIN ABD RAHMAN	AI230086
3	RUSDI BIN ABD RASHID	AI230025
4	MUHAMAD LUKHMAN HAKIM BIN MOHD NIZAM	CI230009
5	MUHAMAD SYAH FIRDAUS BIN AB MANAH	AI230078

UNIVERSITI TUN HUSSEIN ONN MALAYSIA

Faculty of Computer Science and Information Technology

Table of Contents

Introduction..... 3
Project Scope.....3
Network Requirement Analysis
Physical Network Design
Logical Network Design
Network configuration.....3
Configure with Access Control List (ACL) and Demilitarized Zone (DMZ) Area
Network Testing and verification procedure..... 3
Bill of material..... 3

Introduction

XYZ Corp. is a rapidly growing e-commerce company that provides online sales services to a wide range of customers. As the company expands, the existing network infrastructure must be enhanced to support higher traffic volume, ensure service availability, and protect sensitive business and customer data. E-commerce platforms are highly exposed to cyber threats such as unauthorized access, data breaches, and denial-of-service attacks due to their continuous connection to the internet. Therefore, a secure and scalable network infrastructure is essential to maintain business continuity and customer trust. This project focuses on designing and implementing a secure network architecture using a Cisco ASA firewall, internal switches, workstations, and servers, with the goal of improving network security, performance, and reliability. The distribution of work is shown in table 1.

Table 1

Name	Role
Alkhattab	Project Manager
Rusdi	Network Implementation Engineer
Syazwan	Network Designer
Lukhman	Testing and Security Analyst
Syah Firdaus	Documentation and Presentation

Project Scope

The scope of this project covers the design and implementation of a secure network infrastructure for XYZ Corp. The project includes configuring a Cisco ASA firewall to control and monitor all inbound and outbound network traffic, ensuring that external access to the company's network is properly secured. A Demilitarized Zone (DMZ) will be implemented to host public-facing services such as a web server and an email server, allowing customers to access these services without exposing the internal network. Additionally, the internal Local Area Network (LAN) will be designed to enable secure communication among employee workstations and internal resources. The project also involves integrating two network switches to connect internal PCs and ensure efficient data transmission within the organization. This project does not include application-level security or cloud infrastructure but focuses solely on network-level security and architecture. These scopes can be summarized as:

- Design and implementation of a secure network infrastructure for XYZ Corp.
- Configuration of a Cisco ASA firewall for secure external internet access.
- Implementation of a DMZ to host public services such as web and email servers.
- Design of a secure internal LAN for employee communication.
- Integration of two switches to connect internal PCs and devices.
- Focus on network-level security; application and cloud security are excluded.

Network Requirement Analysis

The company's network infrastructure requires specific hardware components to ensure security, efficiency, and scalability. Table 2 shows the following hardware that has been selected based on the network's requirements:

Table 2

Hardware	Justification
Cisco ASA Firewall	Provides secure external access by filtering incoming and outgoing traffic, enforcing NAT, and implementing Access Control Lists (ACLs) to protect internal resources from cyber threats.
Switch1 (Layer 2)	Connects internal devices such as PC1 and PC2, ensuring fast and reliable communication within the internal LAN.
Switch2 (Layer 2)	Connects public servers (Web Server and Email Server) in the DMZ, isolating them from the internal network while allowing external access.
PC1 & PC2	Employee workstations that need secure access to internal resources and DMZ-hosted services.
Web Server	Hosts the company's website, providing public access to customers.
Email Server	Manages company email services, allowing secure communication with clients and internal users.
Router	Connects the internal network to the internet via the firewall, enabling external connectivity while routing traffic appropriately.
Network Cables & Accessories	Physical connections between devices, ensuring reliable communication and network performance.

The network requires devices that can manage traffic, protect internal resources, and support public-facing services. A Cisco ASA firewall is essential to secure external internet access, enforce access control lists (ACLs), and implement network address translation (NAT) for safe communication with the internet. Two Layer 2 switches are required: Switch1 connects employee PCs within the internal LAN, enabling fast and secure internal communication, while Switch2 manages the DMZ segment, hosting public servers such as the Web Server and Email Server, and isolating them from internal resources. PC1 and PC2 serve as employee workstations needing access to internal and DMZ-hosted services. The Web Server hosts the company's website, and the Email Server manages email communications, both of which must be accessible externally but remain secure from internal network threats. A router connects the firewall to the internet, and network cables and accessories ensure reliable physical connectivity between devices.

Physical Network Design

This section discusses the physical topology of the network. As shown in **Figure** 1 and **Figure** 2.

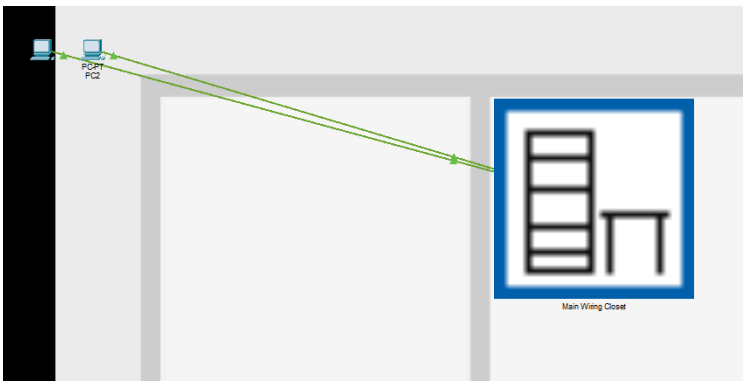


Figure 1

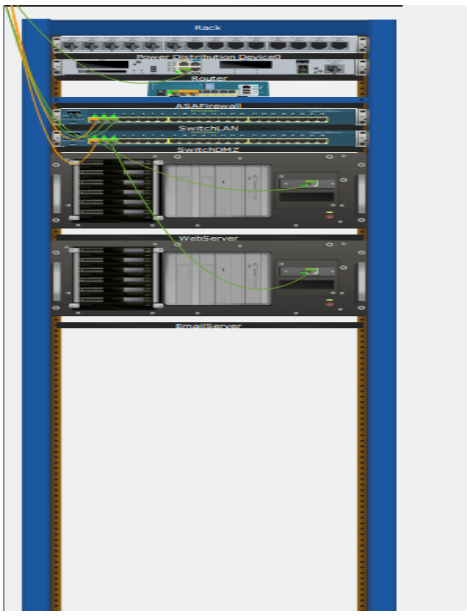


Figure 2

The physical network is centered around a Main Wiring Closet that houses all core equipment in an organized setup. A standard 19-inch rack holds the **Cisco 1841 Router**, **ASA 5505 Firewall**, and two **2950-24 Switches**, arranged to save space and allow proper airflow. A dedicated power distribution device at the top of the rack ensures all critical devices have a reliable power source.

Connections extend from the rack to the office using horizontal cabling. **PC1 and PC2** are located in a separate office area and connect back to **SwitchLAN** through high-speed Ethernet. Inside the rack, patch cables link the switches to the **Web Server** and **Email Server**, which are connected to **SwitchDMZ**. This rack-based setup centralizes the hardware, improves security, protects devices from hazards, and makes maintenance easier by keeping all cables and devices in one accessible location.

Logical Network Design

Network configuration

Configure with Access Control List (ACL) and Demilitarized Zone (DMZ) Area

Network Testing and verification procedure

This section outlines the comprehensive testing procedures conducted to validate the functionality, security, and reliability of the proposed network design. The primary objective of this verification phase is to ensure that all network services—including Web, Email, and SSH remote management are accessible to authorized users while strictly enforcing security policies that protect internal resources.

The testing methodology is divided into different distinct zones, each simulating a specific user perspective:

a. **Inside Network Test:** Verifies that internal staff can securely manage network devices, access local DMZ resources, and browse the external Internet.
b. **DMZ Test:** Confirms the isolation of the Demilitarized Zone, ensuring that compromised public-facing servers cannot initiate unauthorized connections to the secure internal network.

Inside Test:

1. **Internal LAN Connectivity**

```
Pinging 192.168.120.10 with 32 bytes of data:

Reply from 192.168.120.10: bytes=32 time=21ms TTL=127
Reply from 192.168.120.10: bytes=32 time=8ms TTL=127
Reply from 192.168.120.10: bytes=32 time=4ms TTL=127
Reply from 192.168.120.10: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.120.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 21ms, Average = 8ms
```

A standard ICMP echo request was performed between two internal host within the same subnet. As the packet were set and received successfully with 0% packet loss, a successful connectivity was achieved. This confirms that the Layer 2 Switch configuration (VLANs) is functional and that internal hosts can communicate locally without being blocked by the firewall.

2. **Inside to DMZ connectivity**

```
Pinging 192.168.120.10 with 32 bytes of data:

Reply from 192.168.120.10: bytes=32 time=21ms TTL=127
Reply from 192.168.120.10: bytes=32 time=8ms TTL=127
Reply from 192.168.120.10: bytes=32 time=4ms TTL=127
Reply from 192.168.120.10: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.120.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 21ms, Average = 8ms
```

Internal PCs has successfully pinged the web server and email survey. PC can also successfully accessed the servers homepage via web browser. As the ICMP replies were received by the PC, the web pages loaded successfully verifying tht the ASA firewall allows traffic to flow from a high security level (Inside level 100) to Medium security level (DMZ level 50). It also proves the HTTP service on the server is active and the routing path between subnets is valid.

3. Internet Access Verification (Inside to Outside)

```
C:\>ping 205.1.50.121

Pinging 205.1.50.121 with 32 bytes of data:

Request timed out.
Reply from 205.1.50.121: bytes=32 time=1ms TTL=254
Reply from 205.1.50.121: bytes=32 time<1ms TTL=254
Reply from 205.1.50.121: bytes=32 time=1ms TTL=254

Ping statistics for 205.1.50.121:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 205.1.50.121

Pinging 205.1.50.121 with 32 bytes of data:

Reply from 205.1.50.121: bytes=32 time=21ms TTL=254
Reply from 205.1.50.121: bytes=32 time=1ms TTL=254
Reply from 205.1.50.121: bytes=32 time=1ms TTL=254
Reply from 205.1.50.121: bytes=32 time=1ms TTL=254

Ping statistics for 205.1.50.121:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 21ms, Average = 6ms
```

A ping test was conducted from internal PC to ISP's router interface, which result in a success of receiving a reply at 0% packet loss. This confirms the Dynamic PAT is functional on the ASA. The

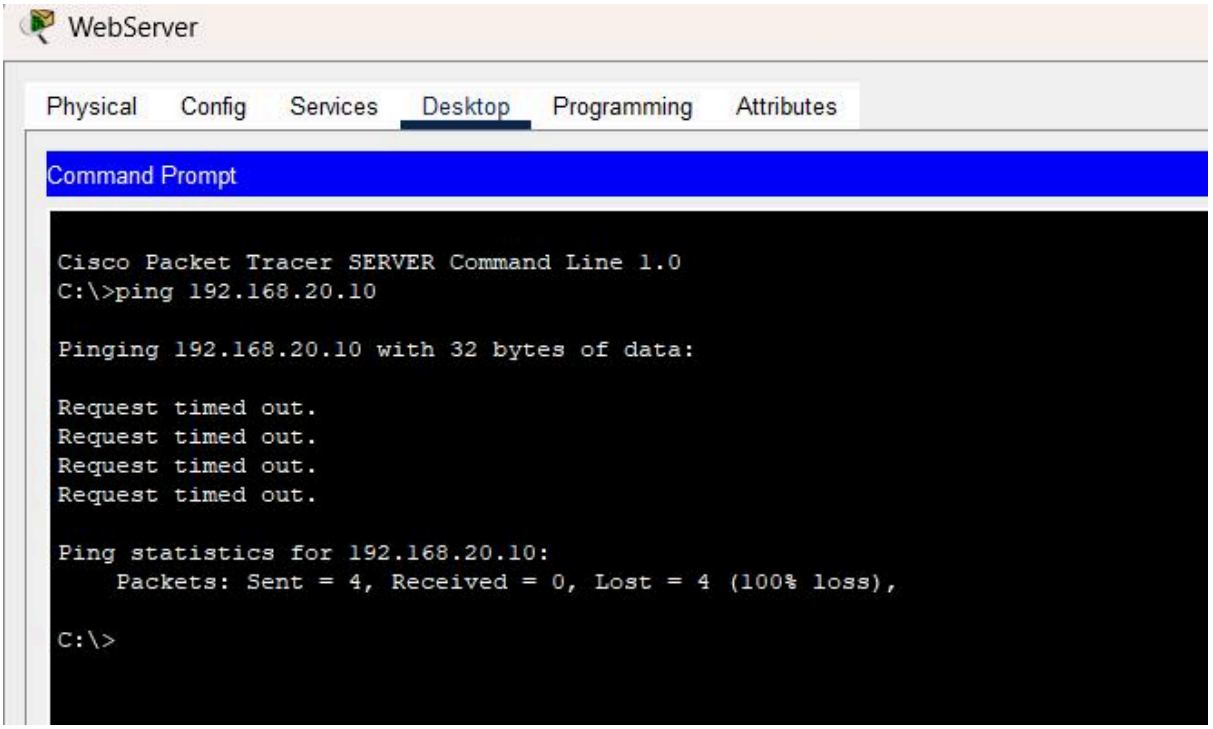
internal private IP addresses are successfully being translated to the public-facing interface IP, allowing users to reach external resources while hiding their internal identities.

4. **Remote management access (SSH)**

```
C:\>ssh -l admin 192.168.20.120

Password:
% Login invalid


Password:



ciscoasa>
```

PC1 successfully establishes an SSH connection to the ASA's internal IP. The connection was established after generating RSA keys and defining local user credentials. This proves that the firewall is securely configured to allow management access only from authorized internal segments.

DMZ Test :



Web server attempted to ping PC1 in the DMZ to inside connectivity test. As shown in the image, the test result in 100% packet loss, showing a successful security result. This result confirms that the Cisco ASA Firewall is correctly enforcing security levels. Traffic is being blocked because the DMZ (Security Level 50) is lower than the Inside (Security Level 100).

Bill of material

The following section details the Bill of Materials (BOM) required for the network deployment. This table itemizes key infrastructure components, including the **Cisco ASA 5505** for perimeter security, **Cisco 1841** routers, and **Cisco 2950** switches. It also accounts for essential end devices such as **HP Pro SFF** PCs and dedicated servers. The breakdown provides a clear financial overview, summing the unit costs of all active equipment and cabling to arrive at a total project estimate of **RM 13,764**. The table below shows a detailed bill of material receipt:

Item	Model / Brand	Qty	Unit Price (MYR)	Total (MYR)
Cisco ASA Firewall	ASA 5505	1	RM 1510	RM 1510
Router	Cisco 1841	1	RM 329	RM 329
Switch	Cisco 2950-24	2	RM 450	RM 900
Employee PC	HP Pro SFF 400 G9	2	RM 2450	RM 4900
Servers	Server-PT	2	RM 3000	RM 6000
Ethernet Cable	Cat 6	5	RM 25	RM 125
Estimated Total Cost: RM 13,764				

Table 00: Bill Of Material